PROBLEM SET 2

FLOATING POINT FORMAT and r and (r-1) complements

1. Convert the decimal number 3.248×10^4 to a single-precision floating-point binary number.
2. Determine the decimal value of the following floating-point binary number:
   0 10011000 10000100010100110000000
3. Express each of the following sign-magnitude binary numbers in single-precision floatingpoint format:
   (a) 0111110000101011 (b) 100110000011000
4. Determine the values of the following single-precision floating-point numbers:

   (a) 1 10000001 01001001110001000000000

   (b) 0 11001100 10000111110100100000000

5. Subtract the following decimal numbers by the 9's and 10's complement methods.

(a) 274- 86 (b) 93- 615 (c) 574.6 – 279.7 (d) 376.3-765.6

6. Find the 12-bit 2's complement form of the following decimal numbers.

   (a) - 37 (b) - 173 (c) – 65.5 (d) – 197.5

7. . Find the 12-bit 1's complement form of the following decimal numbers.

   (a) -97 (b) -224 (c) -205.75 (d) -29.375

8. Subtract the following decimal numbers using 12-bit 2's complement arithmetic.

   (a) 46 – 19 (b) 27 – 75 (c) 125.25- 46.75 (d) 3675- 89.5

9. Subtract the following decimal numbers using the 12-bit 1's complement arithmetic.

   (a) 52 – 17 (b) 46 – 84 (c) 63.75 - 17.5 (d) 73.5 – 112.75

10. Represent the following decimal numbers in 8-bit : (i) sign magnitude form (ii) sign 1's complement form, and (iii) sign 2's complement form.

(a) +14 (b) +27 (c) +45 (d) -17 (e) -37 (f) -76